ADVANCED DATA SCIENCE (ADS)

End-Term Project Guidelines and Three-Stage Presentation Framework

An evidence-driven data science workflow

1. Project Objectives

The End-Term Project carries **40% of the overall assessment**, distributed as follows:

Component	Marks	Share
Pre-submission	10	10%
Final report	10	10%
Final presentation	20	20%
Total	**40**	**40%**

The project must demonstrate a complete data science workflow, from problem definition and data acquisition to analysis, hypothesis validation, and actionable recommendations.

Projects must follow the **BADIR framework** and address a meaningful business or policy problem using methods prescribed in the ADS course handout.

2. Dataset and Data-Source Requirements

Requirement	Minimum / Rule	Detail
Dataset size	≥ 100,000 rows	In the final integrated dataset.
Number of variables	≥ 20 (40+ preferred)	Aim for 40 or more wherever feasible.
Data sources	No Kaggle	Use multiple credible sources, preferably at least two independent ones, such as government portals, APIs and research repositories.

- **Data authenticity:** Variables must represent genuine, observed or recorded attributes. Derived variables must be transparently calculated from authentic data.
- **Data provenance:** Document source links, collection dates, time coverage, and relevant metadata.
- **Single Source of Truth (SSOT):** Integrate the sources into one consistent dataset. Explain join keys, integration logic, units, schema, duplicate handling, missing-value treatment, and data-quality checks.
- **Course methods:** Apply all relevant methods prescribed in the ADS course handout. Do not go beyond the prescribed course scope.
- **BADIR framework:** Use BADIR to structure the problem-solving process and project deliverables.
- **Evidence and validation:** Support major findings using complementary forms of evidence, such as statistical tests, visualisations, robustness checks, and independent-source comparisons, where appropriate.
- **Limitations and deviations:** Document data constraints and obtain instructor approval for deviations from mandatory requirements.

3. Three-Stage Presentation Framework

Stage 1: Stakeholder Discussion and Problem Definition

Recommended: 3–5 slides

#	Slide	Content
1	**Stakeholder and Problem**	Keep the stakeholder at the centre of the discussion. Understand their needs, challenges, objectives, and decision-making context before defining the business or policy problem.
2	**Research Questions and Hypotheses**	Formulate research questions and initial hypotheses based on the stakeholder's needs, assumptions, and expected outcomes.
3	**Data Audit**	Identify potential data sources, variables, data volume, quality, time coverage, and access constraints.
4	**Data Integration and SSOT Plan**	Explain how data from multiple sources will be linked, harmonised, and integrated into a consistent dataset.
5	**Project Plan**	Map the workflow, prescribed course methods, milestones, and deliverables to the BADIR framework.

Stage 2: Data Analysis and Hypothesis Validation

Recommended: 3–5 slides

#	Slide	Content
1	**Data Preparation**	Summarise data integration, cleaning, missing-value treatment, outlier handling, and quality checks.
2	**Exploratory Data Analysis (EDA)**	Present relevant distributions, trends, relationships, and visualisations.
3	**Hypothesis Testing**	Report the methods used, supporting evidence, interpretations, and hypothesis status.
4	**Unsupported or Inconclusive Hypotheses**	Explain contradictory evidence, insufficient data, or measurement limitations.
5	**Revised Analysis Plan**	Identify remaining analyses, refinements, and unresolved questions.

Stage 3: Final Inference and Decision Impact

Recommended: 3–4 slides

#	Slide	Content
1	**Final Findings**	Summarise the key conclusions and classify hypotheses as supported, not supported, or inconclusive.
2	**Business and Policy Implications**	Connect the findings to stakeholder decisions and propose actionable recommendations.

#	Slide	Content
3	**Impact and Limitations**	Discuss plausible benefits, uncertainty, assumptions, and analytical constraints.
4	**Implementation Roadmap**	Propose prioritised actions, timelines, responsible stakeholders, and measurable indicators, where appropriate.

4. Expectations Across All Stages

- Maintain a clear link between the stakeholder's needs, problem definition, research questions, hypotheses, data, methods, findings, and recommendations.
- Meet the minimum dataset requirements and document data provenance and integration.
- Use authentic data from multiple credible sources. Kaggle datasets are prohibited.
- Demonstrate a coherent SSOT rather than simply combining unrelated files.
- Apply the prescribed course methods and support major findings with multiple lines of evidence wherever appropriate.
- Treat unsupported or inconclusive hypotheses as valid outcomes when justified by evidence.
- Distinguish association from causation and avoid overstating business or policy implications.
- Transparently report assumptions, transformations, exclusions, and limitations.
- Keep presentations concise, readable, evidence-driven, and focused on decisions.

5. Overall Project Progression

Stage 1: Define the Problem

Understand the stakeholder's needs and challenges, formulate research questions and hypotheses, audit data sources, and plan data integration.

Stage 2: Validate the Evidence

Prepare and integrate the data, conduct EDA, apply prescribed methods, and assess hypotheses.

Stage 3: Inform the Decision

Present defensible conclusions, explain implications for the stakeholder, acknowledge limitations, and recommend actionable measures.

> **Core principle:** Understand the stakeholder → Define the problem → Validate the evidence → Inform the decision.

6. Final Submission Requirements

The final submission must include:

1. Problem statement, research questions, and hypotheses.
2. Data sources, provenance, and integration methodology.
3. Final integrated dataset or a reproducible procedure for accessing it.
4. Data-quality checks and EDA.
5. Applicable course methods and hypothesis assessments.
6. Supporting evidence, final findings, and limitations.

7. Business or policy recommendations and an implementation roadmap.
8. Reproducible code and supporting documentation, where applicable.

> **Evaluation principle:** Project quality will be assessed on analytical rigour, data authenticity, integration quality, transparency, evidence-based reasoning, stakeholder relevance, and the practical value of conclusions, not simply on the number of hypotheses supported.